SUR-TRAC-E

SURVEILLANCE ET TRACKING DE L'ESPACE



REAL-TIME SPACE SURVEILLANCE

- Latency
- Limited coverge
- Data processing constraints



SPACE THREATS

SUSPICIOUS SATELLITE MANEUVER



OUR SOLUTION

SUR-TRAC-E

ONBOARD INFRASTRUCTURE

- 4 satellites per orbital plane
- 24 U each
- LWIR-RF-Hyperspectral
- On - board AI processing
- Alert system



DEFEND STRATEGICAL INFRASTRUCTURES

DETECT

Real time anomaly

COMPUTE

and AI Data processing

ALERT

the client in real-time

FUTURE MILITARY OPERATION COORDINATION IN ORBIT



ALERT SYSTEM

01

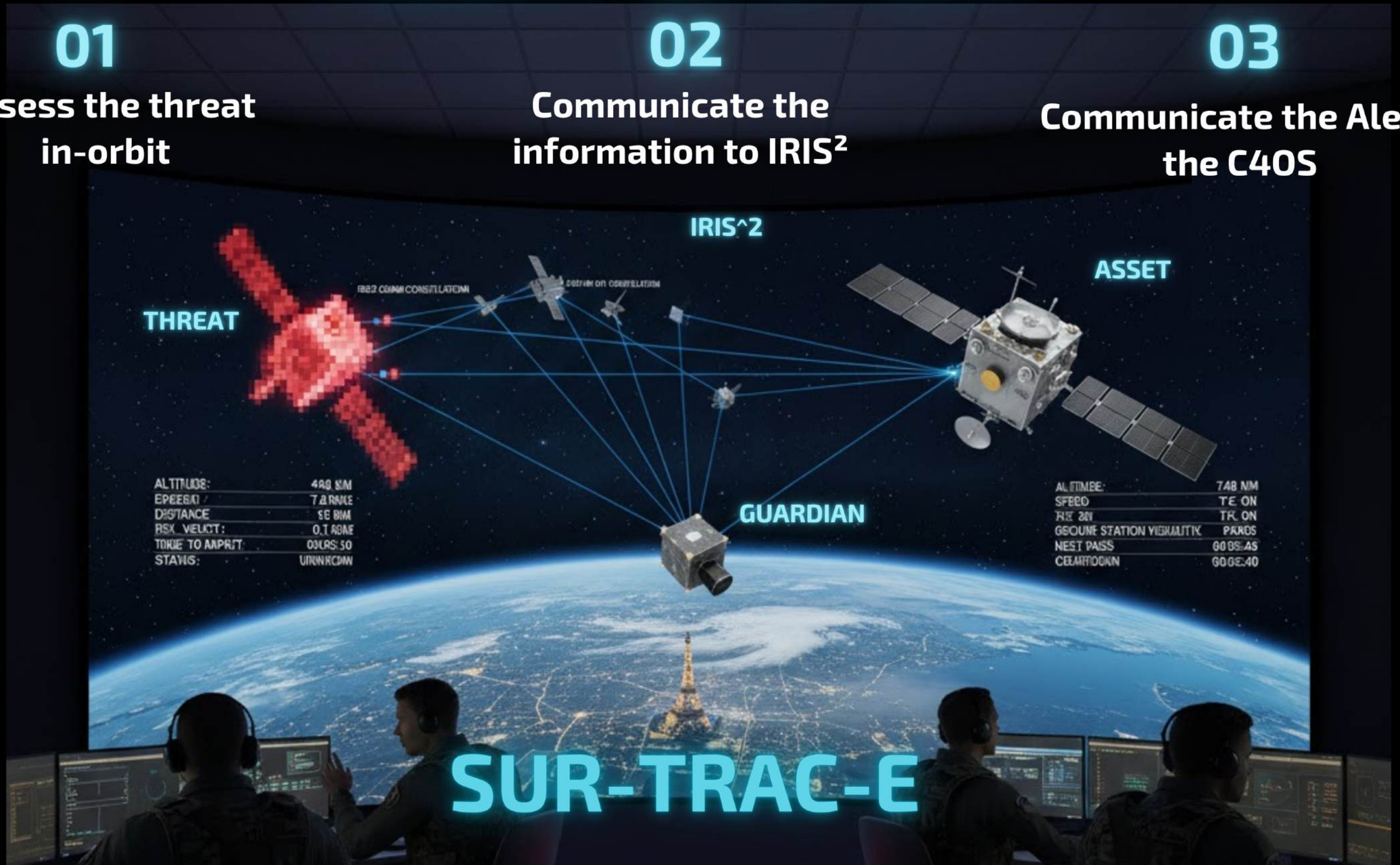
Assess the threat in-orbit

02

Communicate the information to IRIS²

03

Communicate the Alert the C4OS



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SWOT ANALYSIS



STRENGTHS

- Multi-modal sensing capability
- **On-board AI and real-time analysis**
- Real-time anomaly detection and alerting
- **Interoperability with existing databases**



WEAKNESSES

- **Limited on-board power and processing resources**
- Communication bandwidth and latency
- Dependence on external catalogs and networks



OPPORTUNITIES

- **European defense and space security initiatives**
- Dual-use civilian and military applications
- Advancement of on-board AI autonomy
- Data commercialization and analytics services



THREATS

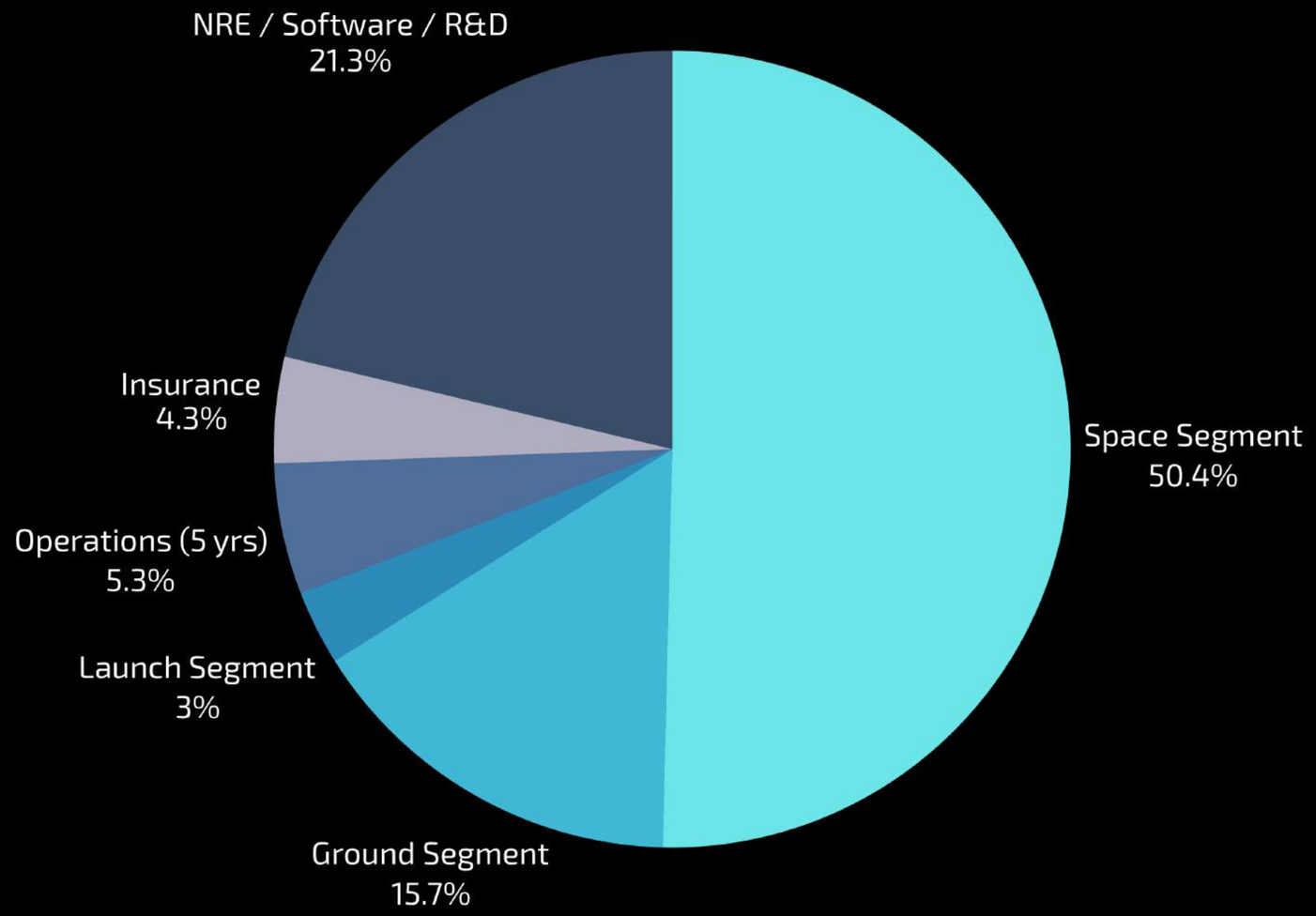
- Cybersecurity and data integrity risks
- Competition from U.S. and Chinese constellations
- **AI decision transparency and accountability**

ROADMAP TIMELINE



COST ESTIMATION

Expected Total 39.5 MLN €



TEAM



Maria Mattiello
Payload Engineer



Pierluca De Felice
OBC Engineer



Raffaele Colamarino
System Engineer



Vincenzo Jr Di Rosa
Strategic Researcher



Corrado D'Urso
MA Engineer

QUESTIONS?